

When Simple Wins: Leakage-Free Benchmarking of Classical Machine Learning vs. Graph Neural Networks in Low-Data Bioactivity Prediction^{*}

Andrey Vinajera-Zamora^a, Samuel Montejó-Sánchez^{a,*}, Juan Castillo-Garit^a and Raul Arias-Carrasco^a

^a*Instituto Universitario de Investigación y Desarrollo Tecnológico, Universidad Tecnológica Metropolitana, Ignacio Valdivieso 2409, Región Metropolitana, 97279, Santiago, Chile*

^b*Departamento de Informática y Computación, Universidad Tecnológica Metropolitana, José Pedro Alessandri 1242, Ñuñoa, Región Metropolitana, 7750000, Santiago, Chile*

ABSTRACT

Predictive modeling in low-data bioactivity regimes faces significant methodological challenges, including high susceptibility to overfitting, silent data leakage, and unwarranted architectural complexity. Here, we present a leak-free benchmarking framework evaluating classical machine learning stacking ensembles against state-of-the-art Graph Neural Networks (ChemProp D-MPNN) for the inhibition of *Toxoplasma gondii* dihydrofolate reductase (TgDHFR, $N = 756$). Utilizing a rigorous 15-fold nested cross-validation protocol, we systematically compared four molecular representation spaces spanning 2D topological fingerprints, 2D physicochemical descriptors, and 3D geometric conformers. Classical stacking ensembles trained on lightweight Morgan fingerprints achieved superior predictive performance ($R^2 = 0.7407 \pm 0.0531$, MAE = 0.4830), statistically matching the performance of computationally intensive 2D/3D hybrid descriptors ($R^2 = 0.7387$, $p = 0.8183$) while entirely eliminating the conformational overhead of 3D modeling. Conversely, the D-MPNN underperformed ($R^2 = 0.6395$, MAE = 0.5841), demonstrating that data scarcity restricts end-to-end representation learning. Deploying the optimized classical model to screen FDA-approved therapeutics prioritized high-potency antifolates alongside high-efficiency scaffolds, including triamterene (ligand efficiency = 0.49) and pyrimethamine (ligand efficiency = 0.52). Subsequent orthogonal validation via AutoDock Vina molecular docking resolved screening discordances, successfully filtering out false positives such as bisacodyl while confirming high-affinity consensus leads, most notably Pyrimethamine (docking energy: -7.31 kcal/mol). These findings demonstrate that in data-constrained QSAR regimes, the explicit inductive bias of fixed structural fingerprints acts as a natural regularizer that outperforms deep graph representations, providing a reproducible and computationally efficient paradigm for target-specific drug repurposing.

Keywords: Quantitative Structure-Activity Relationship, Low-Data Regimes, Nested Cross-Validation, Data Leakage, Graph Neural Networks, Molecular Docking

1. Introduction

1.1. Predictive Modeling in Low-Data Bioactivity Regimes

It is a well-established dogma in quantitative structure-activity relationships (QSAR) that parsimonious models generalize best Winkler (2026). In data-limited scenarios, simpler interpretable approaches like multivariate linear regression have traditionally prevailed, often demonstrating prediction reliability

^{*} smontejo@utem.cl

 avinajera@utem.cl (A. Vinajera-Zamora); smontejo@utem.cl (S. Montejó-Sánchez); j.castillo@utem.cl (J. Castillo-Garit); raul.arias@utem.cl (R. Arias-Carrasco)

 <https://orcid.org/0000-0002-4922-457X> (A. Vinajera-Zamora); <https://ieeexplore.ieee.org/author/37085682270> (S. Montejó-Sánchez); <https://orcid.org/0000-0003-0896-9484> (J. Castillo-Garit); <https://orcid.org/0000-0003-1698-9282> (R. Arias-Carrasco)
ORCID(s):

equivalent to complex artificial neural networks Shalit Peleg and Milo (2023); Xie, Wiriyarattanakul, Rungrotmongkol, Shi, Wiriyarattanakul and Maitarad (2023). These linear models maintain higher bias and lower variance, avoiding the extreme sensitivity to random fluctuations seen in complex algorithms, especially when supplemented by rigorous quantum chemical calculations to derive robust descriptors Xie et al. (2023).

Modeling small datasets is inherently susceptible to overfitting, a bottleneck severely exacerbated when the high dimensionality of molecular descriptors vastly exceeds the sample size Hawkins (2004); Goodarzi, Heyden and Funar-Timofei (2013). To combat this data sparsity, researchers frequently implement data augmentation (DAU) or synthetic sample generation to expand the training space Gerz and Jelali (2026); Meyer, Andreeva, Weiss, Ding, Shen, Wang, Chen and Dagnew (2025). However, these techniques introduce non-negligible computational complexity and resource demands, often requiring high-performance hardware and iterative optimization cycles Meyer et al. (2025); Calle, Bates, Reynolds, Liu, Cui, Ly, Wang, Zhang, de Armendi, Shettar, Fu, Tang and Pan (2025). More critically, while DAU is designed to enhance model learning, its improper application often masks the true efficacy of the base algorithms by artificially elevating performance metrics through latent data dependencies and process-induced leakage Gerz and Jelali (2026); John, Saurette and Heung (2025).

Applying complex, non-linear architectures to these regimes introduces profound validation hurdles. When augmentation is applied globally before dataset partitioning ("augment-then-split"), it violates core independent and identically distributed (i.i.d.) assumptions, allowing models to memorize augmentation artifacts rather than learn genuine physical rules Gerz and Jelali (2026); Veetil, Chowdary, Chowdary, Sowmya and Gopalakrishnan (2024). This danger extends directly into model selection; tuning hyperparameters is exceptionally prone to overfitting when sample sizes are small and the parameter space is large Cawley and Talbot (2010). Consequently, evaluating machine learning architectures in isolation is meaningless without rigorously assessing the combined model selection procedure, making double (or nested) cross-validation mandatory to ensure true independence between model building and selection Baumann and Baumann (2014); Cawley and Talbot (2010).

Traditional evaluation metrics and standard k-fold cross-validation often misleadingly overestimate explorative prediction capabilities Király, Kiss, Kovács, Ballaj and Tóth (2022); Xiong, Cui, Liu, Zhao, Hu and Hu (2020). Data augmentation strategies can exacerbate underlying data structures, yielding overly optimistic accuracy metrics that mislead stakeholders regarding the model's true explorative power John et al. (2025). Furthermore, algorithms like Random Forest fundamentally lack extrapolation capability Xiong et al. (2020), and generative deep learning faces severe limitations with sparse, discrete chemical spaces like SMILES strings Liu, Liu, Duan, Wang, Zhang and Li (2023). Therefore, to safeguard methodological integrity, any generative or augmentative process must be confined strictly to the training portion of the folds after a rigorous, independent partition has been established Gerz and Jelali (2026); John et al. (2025).

To safely harness the potential of overparameterized models Winkler (2026) without falling into the overfitting trap, rigorous validation protocols are mandatory. Techniques such as y-randomization (response shuffling) become essential to verify that the model captures true chemical relationships rather than random noise Rücker, Rücker and Meringer (2007). Conversely, when descriptors are meticulously curated and validated, machine learning can effectively guide rational design, successfully discriminating potent bioactives—such as Amyloid- β aggregation inhibitors—from decoys Bansal, Sharma, Chakraborty, Sunkaria and Jain (2025).

Methodologies like transfer learning bypass data scarcity by adapting base networks previously optimized on large source domains Espley, Farrar, Buttar, Tomasi and Grayson (2023). By utilizing reduced learning rates during fine-tuning Bodun, Ibrahim, Bashiru, Abolade, Ndarawit, John-Joy Owolade, Temitope, Somtochukwu, Olugbogi, Samadhi, Oduselu, Sharif, Omoboyowa and Ajani (2025), and preferring efficient architectures like convolutional neural networks over heavier Transformer models Özçelik and Grisoni (2024), researchers can robustly navigate class imbalance and model selection bias in low-resource settings.

1.2. Domain-Engineered Features versus End-to-End Representation Learning

Traditional feature-based machine learning relies heavily on domain-engineered descriptors and molecular fingerprints Das, Sinnarasan, Paul, Sheikh and Venkatesan (2026). Methods like Morgan (ECFP4) encode structural traits, while topological, 3D, and physicochemical descriptors—such as radial distribution

functions, Burden eigenvalues, and ab initio quantum chemical calculations—capture specific molecular interactions deterministically without requiring massive data Bansal et al. (2025); Iwuala, Madumelu, Aliyu, Ayo, Hamza and Habila (2026); Xie et al. (2023). This approach embeds theoretical chemistry directly into the model, ensuring high interpretability and clarity Xie et al. (2023). However, accumulating these handcrafted features creates highly dimensional, redundant matrices Goodarzi et al. (2013); Shalit Peleg and Milo (2023). In small datasets, this redundancy increases computational costs and severely worsens the feature selection bottleneck Goodarzi et al. (2013); Shalit Peleg and Milo (2023).

Alternatively, end-to-end representation learning bypasses manual feature selection by autonomously mapping discrete molecular topologies into continuous latent spaces Basit Iqbal, Assad, Bhat, Macha and Shah (2025); Azad (2026). Deep learning paradigms—including Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Graph Neural Networks (GNNs)—extract task-specific traits directly from chemical structures to model complex, nonlinear interactions Azad (2026); Das et al. (2026); Liu et al. (2023). Similarly, Chemical Language Processing (CLP) applies natural language techniques to string representations like SMILES Özçelik and Grisoni (2024). However, learning representations in discrete chemical spaces is intrinsically fragile; altering even a single character in a SMILES string can irreparably invalidate the entire molecular structure Liu et al. (2023).

Despite these advantages, the utility of deep learning complexity in low-data regimes remains highly contested. These algorithms are inherently data-hungry, making representation learning highly ineffective when ground-truth labels and dataset volumes are severely restricted Konstantakos, Cani, Mademlis, Chalkiadaki, Asano, Gavves and Papadopoulos (2025). To circumvent this limitation in the specific domain of *Toxoplasma gondii* dihydrofolate reductase (TgDHFR) inhibition, recent studies have resorted to highly complex pipelines. For instance, Karimova and Senanayake (2025) employed Gaussian noise-based data augmentation and deep neural network (DNN) ensembles to stabilize training on a restricted dataset, elevating their baseline R^2 from 0.75 to 0.85. This raises a fundamental research question: is such statistical complexity and synthetic data generation strictly necessary to achieve competitive predictive reliability, or can a parsimonious classical machine learning ensemble achieve equivalent baseline performance ($R^2 \approx 0.75$) “out-of-the-box” using standard domain-engineered features? Because small datasets cannot naturally sustain deep architectures without risking catastrophic overfitting Shalit Peleg and Milo (2023); Dar, Şaban Öztürk, Özbey, Oguz and Çukur (2023); Espley et al. (2023), simpler classical ensembles frequently achieve equivalent or superior predictive reliability in data-scarce scenarios Iwuala et al. (2026); Das et al. (2026); Xie et al. (2023).

Nevertheless, traditional skepticism toward nonlinear models in limited-data scenarios is changing Dalmau, Sigman and Alegre-Requena (2025). When properly tuned, regularized, and hybridized with structured linear priors or heuristics to stabilize training, nonlinear algorithms can achieve competitive results comparable to robust classical methods Dalmau et al. (2025); Dar et al. (2023). Furthermore, transfer learning serves as a vital catalyst. By fine-tuning pretrained networks like ChemBERTa, researchers can achieve accurate representation learning predictions even with only dozens of target data points Oulladji, Saadallah, Guellil, Abbad and Bekhti (2025); Espley et al. (2023).

1.3. Methodological Rigor and the Threat of Data Leakage in Molecular Modeling

Data leakage—the unintended introduction of test data information into the training process—is a statistically silent error that fundamentally undermines the validity of machine learning evaluations by inflating metrics without triggering code warnings Vrba, Steinbach, Jirsa and Homma (2026); Gerz and Jelali (2026). In tabular molecular descriptor pipelines, this phenomenon typically manifests when global feature scaling (e.g., z-score standardization or MinMax scaling), descriptor imputation, or feature-filtering procedures (such as variance-threshold filtering or supervised feature selection) are executed prior to fold partitioning Correa, Tavakoli, Vogel and Gebbers (2025); Lewis, Spiliopoulou, Goldmann, Pitzalis, McKeigue and Barnes (2023). Applying scaling globally calculates dataset-wide statistics, such as the mean and standard deviation, which inadvertently leaks information regarding the test set distribution into the training process. Consequently, this violates the fundamental independence assumption between partitions and yields overoptimistic, biased performance estimates Cawley and Talbot (2010). Similarly, executing feature selection or variance-based filtering over the entire dataset introduces a selection bias, as the chosen predictors retain a partial “memory” of the hold-out test data Varma and Simon (2006).

To eliminate these vulnerabilities, preprocessing, scaling, and feature selection must be treated as integral components of the model-fitting workflow Cawley and Talbot (2010). All transformation parameters must be strictly computed within each inner training fold and subsequently applied to the validation and test folds without parameter re-estimation Vrba et al. (2026). Rigorous partitioning protocols, such as strict three-way splits, must also be enforced to guarantee absolutely no overlap among training, validation, and test sets across subjects or discrete samples Dar et al. (2023).

Beyond preprocessing, data leakage frequently corrupts the model selection phase. Tuning hyperparameters globally across the entire dataset grants the algorithm illicit knowledge of the test distribution, yielding spuriously high performance Veetil et al. (2024). Even in traditional internal validation, an overlap between the data used for hyperparameter tuning and the test set creates an optimistic bias that renders the evaluation uninformative regarding real-world generalizability John et al. (2025). The validation dataset fundamentally loses its independence if its collective predictions continuously influence the search for an optimal model, triggering severe model selection bias. This lack of independence causes models—especially highly parameterized deep learning architectures deployed on small pretraining datasets Konstantakos et al. (2025)—to overfit by adapting to noise, producing deceptively overoptimistic internal metrics that severely underestimate true generalization errors Baumann and Baumann (2014).

To mitigate these structural biases, model selection must operate as an encapsulated component of the fitting process. Double or nested cross-validation is mandatory to generate strictly independent test objects in an outer loop that remain entirely isolated from model building and hyperparameter optimization Baumann and Baumann (2014). To further prevent fortuitous or biased splitting, researchers can employ deterministic methods like the Kennard-Stone algorithm to construct highly representative, unbiased external test sets Xie et al. (2023). When low-data regimes necessitate data augmentation, methodological purity remains paramount: classifier performance must be measured exclusively on real test data, keeping synthetic samples strictly isolated, while employing probabilistic Bayesian optimization rather than exhaustive grid searches to safely navigate the hyperparameter space Meyer et al. (2025). Preserving and transparently documenting this absolute methodological rigor is critical; insufficient reporting of data partitioning often confounds scientific literature, leading researchers to incorrectly attribute high performance to novel complex architectures when the success actually stems from silent data leakage Rojas and Madanian (2026).

1.4. Objectives and Principal Contributions

This study bridges the gap between stringent statistical validation and advanced representation learning to accelerate targeted drug discovery in low-data regimes.

- **Methodological Reproducibility and Leak-Free Validation:** Implementation of a fully transparent, 100% reproducible Python pipeline built upon a rigorous 15-Fold Nested Cross-Validation protocol. This ensures absolute statistical independence between hyperparameter optimization and model evaluation, effectively eliminating model selection bias and data leakage.
- **Architectural Benchmarking in Data-Limited Regimes:** A comprehensive empirical comparison evaluating the predictive reliability of traditional domain-engineered classical ensembles against state-of-the-art Graph Neural Networks (specifically, the ChemProp Directed Message Passing Neural Network, D-MPNN) using a standardized reference dataset.
- **Empirical Validation of Parsimony over Complexity:** Explicit demonstration that simple classical machine learning ensembles achieve statistical parity ($R^2 \approx 0.75$) with baseline deep learning models reported in recent literature. This establishes that competitive predictive reliability can be attained "out-of-the-box," effectively eliminating the necessity for synthetic data augmentation and computationally expensive 3D geometric preprocessing in low-data regimes.
- **Translational Drug Repurposing Pipeline:** The seamless integration of machine learning predictions with structure-based Molecular Docking. This hybrid pipeline systematically screens, identifies, and prioritizes FDA-approved compounds, providing a validated computational pathway for rapid therapeutic repurposing.

2. Materials and Methods

An overview of the multi-tier virtual screening pipeline implemented in this study is illustrated in Figure 1.

2.1. Bioactivity Benchmark Dataset and Curation

The dataset used in this study was derived from a previously published benchmark on *Toxoplasma gondii* dihydrofolate reductase (TgDHFR) enzyme inhibitors Karimova and Senanayake (2025). The provenance of these data relies on the integration of primary public repositories to ensure chemical and bioactivity representativeness, aligning with foundational QSAR guidelines that mandate rigorous data preparation Tropsha (2010). By utilizing the baseline dataset compiled by Karimova and Senanayake (2025), we establish a strict control framework. This allows us to isolate and evaluate the true predictive power of classical machine learning algorithms against computationally heavy statistical techniques, deliberately omitting synthetic data augmentation to assess performance strictly on empirical ground-truth data.

- **Primary Data Sources:** The initial collection was constructed by combining records from BindingDB, which provided detailed drug-protein interaction parameters and SMILES structures, alongside bioactivity data retrieved from ChEMBL Karimova and Senanayake (2025).
- **Consolidation and Curation Protocol:** Following database integration, automated standardization steps—such as salt stripping and structural validation—were applied to guarantee overall chemical integrity Bodun et al. (2025). Crucially, of the initial 873 entries retrieved, 117 compounds were systematically excluded during this phase to mitigate experimental noise and avoid overfitting Cherkasov, Muratov, Fourches, Varnek, Baskin, Cronin, Dearden, Gramatica, Martin, Todeschini, Consonni, Kuz'min, Cramer, Benigni, Yang, Rathman, Terfloth, Gasteiger, Richard and Tropsha (2013); Iwuala, Aliyu, Ayo, Hamza, Madumelu and Habila (2025). The specific exclusion criteria comprised: (i) the presence of organometallic complexes incompatible with subsequent 3D force-field generation, and (ii) undefined stereocenters that introduce conformational ambiguity.
- **Treatment of Experimental Variations:** To address compounds with multiple IC_{50} measurements obtained under varying conditions, the Interquartile Range (IQR) method was employed. Specifically, duplicate assays exhibiting a bioactivity variance exceeding 1.5 log units were filtered out as statistical outliers Karimova and Senanayake (2025). For the remaining consistent entries, an average IC_{50} value was calculated to establish a single representative measurement per ligand group.
- **Target Variable Formulation:** This rigorous curation protocol yielded a final, highly curated benchmark of $N = 756$ unique small molecules. The continuous target variable was formulated as pIC_{50} , defined as the negative logarithm of the molar IC_{50} concentration ($pIC_{50} = -\log_{10}(IC_{50})$) Karimova and Senanayake (2025). This logarithmic conversion standardizes the response values, reduces variance, and compresses the active range, rendering it optimal for robust regression algorithms.

2.2. Representation Spaces: Molecular Fingerprints and Conformational Descriptors

To systematically evaluate the structure-activity relationship, three distinct molecular representation spaces were programmatically generated using the open-source cheminformatics library RDKit Landrum, Tosco, Kelley, Rodriguez, Cosgrove, Vianello, Gedeck, Jones, Kawashima, Nealschneider et al. (2026):

- **Molecular Fingerprints (ECFP4):** Morgan fingerprints were computed to represent molecular structures as vector spaces Kausar and Falcao (2019). Extended-connectivity fingerprints (ECFPs) were specifically developed for structure-activity modeling because they can be calculated rapidly and represent an essentially infinite number of distinct molecular features, including stereochemical information Rogers and Hahn (2010). Furthermore, classical 2D fingerprint-based models have been demonstrated to perform as well as state-of-the-art 3D structure-based models for bioactivity predictions relying solely on ligand information Gao, Nguyen, Sresht, Mathiowetz, Tu and Wei (2020).
- **2D Physicochemical Descriptors:** A comprehensive set of conventional two-dimensional topological representations was extracted to establish a quantitative physicochemical baseline Kausar and Falcao (2019). Well-curated 2D molecular representations are highly robust, and recent studies have shown

they can maintain scaffold-robust advantages over 3D alternatives in various permeability and property prediction tasks Tiwari, Madra-Gackowska, Gackowski and Szeleszczuk (2026).

- **3D Conformational Descriptors (ETKDG):** Three-dimensional conformers were generated employing the distance geometry approach implemented in RDKit Riniker and Landrum (2015). Because purely distance-based constraints tend to lead to distorted aromatic rings and sp^2 centers, this methodology utilizes experimental torsion-angle preferences obtained from small-molecule crystallographic data to improve conformation generation Riniker and Landrum (2015). Although recent evaluations indicate that 3D conformer descriptors may provide limited incremental predictive value over curated 2D representations in specific tabular benchmark algorithms Tiwari et al. (2026), integrating spatial information remains highly relevant for molecular image-based modeling Matsuzaka and Uesawa (2023) and geometry-augmented representation learning Zhang and Bai (2024).

2.3. Strict 15-Fold Nested Cross-Validation Protocol

In a standard cross-validation protocol, the dataset is partitioned into k disjoint blocks that are systematically rotated between training and testing phases Calle et al. (2025); Krstajic, Buturovic, Leahy and Thomas (2014). While conventional nested cross-validation (NCV) literature frequently employs symmetrical 5×5 or 10×10 architectures Calle et al. (2025); Lewis et al. (2023); Wainer and Cawley (2021), there is no methodological restriction requiring identical fold counts for the inner and outer loops Krstajic et al. (2014). To guarantee leak-free evaluation, balance computational cost, and mitigate model selection bias in this severe low-data regime ($N = 756$), a strict 15×5 asymmetric NCV protocol was implemented. While Bemis-Murcko scaffold splitting is widely used to test out-of-domain extrapolation, a random nested cross-validation scheme was selected here to maintain a consistent bioactivity distribution across folds. This ensures that hyperparameter optimization in the inner loop remains statistically stable without creating underpopulated activity cliffs in the outer test folds. This architecture mathematically decouples model assessment from model selection using two distinct concentric loops:

- **Outer Loop (15-Fold Generalization Assessment):** To achieve a lower variance in performance estimation compared to extreme approaches like Leave-One-Out (LOO), the dataset was partitioned into 15 independent folds Krstajic et al. (2014). In each iteration, one fold is strictly held out as the test set while the remaining 14 folds form the training pool. This ensures the test data is never used to optimize model performance, yielding an unbiased out-of-sample prediction estimate Varma and Simon (2006); Wainer and Cawley (2021).
- **Inner Loop (5-Fold Optimization and Feature Selection):** Within each 14-fold training pool, a secondary 5-fold inner loop was embedded to execute Bayesian hyperparameter optimization (Optuna) and feature selection. Confining these steps entirely within the inner partitions forces the selection of more parsimonious models and prevents inadvertent information leakage from the test set—a critical requirement since filtering predictors on the whole dataset produces severely overoptimistic performance metrics Calle et al. (2025); Lewis et al. (2023); Parvande, Yeh, Paulus and McKinney (2020).

Three distinct machine learning algorithms—Random Forest (RF), eXtreme Gradient Boosting (XGBoost), and Light Gradient Boosting Machine (LightGBM)—were evaluated under this framework. Although computationally intensive, prioritizing this custom nested configuration over a flat cross-validation approach is methodologically mandatory to achieve realistic and robust pIC_{50} predictive estimates Wainer and Cawley (2021); Krstajic et al. (2014).

2.4. Deep Learning Baseline: Directed Message Passing Neural Networks

To benchmark the classical machine learning models against advanced representation learning, a Directed Message Passing Neural Network (D-MPNN) was implemented using the open-source Chemprop framework (version 2.0) in Python Graff, Morgan, Burns, Doner, Li, Li, Manu, Menon, Pang, Wu, Zalte, Zheng, Coley, Green and Greenman (2025); Yang, Swanson, Jin, Coley, Eiden, Gao, Guzman-Perez, Hopper, Kelley, Mathea et al. (2019). To guarantee a mathematically direct comparison with the classical ensembles, the D-MPNN

was evaluated across the exact same 15 outer test folds generated by the nested cross-validation protocol. This alignment removes the partitioning bias inherent to a single hold-out split, ensuring that the resulting metrics accurately reflect out-of-sample generalization on the curated dataset ($N = 756$).

The deep learning architecture processes molecular SMILES strings by transforming them into graph representations using RDKit, where atoms serve as vertices and bonds as edges Heid, Greenman, Chung, Li, Graff, Vermeire, Wu, Green and McGill (2023). The `SimpleMoleculeMolGraphFeaturizer()` constructed initial feature vectors based on atomic identity and topological connectivity. Subsequently, the network applied `BondMessagePassing()` to learn embeddings directly from directed edges rather than vertices. This design prevents spurious feedback loops (totters), significantly reducing noise in the learned representation Yang et al. (2019). These atomic representations were then pooled into a unified molecular vector via a `MeanAggregation()` function.

Prior to training, the continuous pIC_{50} targets underwent internal data normalization. The aggregated embeddings were passed through a `RegressionFFN()` equipped with batch normalization to generate the final predictive output. Model weights were optimized via PyTorch Lightning over a maximum of 100 epochs. This epoch limit, alongside default Chemprop hyperparameters (e.g., hidden size, depth, and learning rate), was deliberately selected to represent a robust baseline. Extensive hyperparameter tuning for deep learning models on datasets of this size frequently induces severe overfitting rather than genuine performance gains Cawley and Talbot (2010); Shalit Peleg and Milo (2023). Therefore, evaluating the network under standard configurations ensures a fair comparison against the baseline capabilities of classical ensembles.

The internal evaluation metric configured during training was Root Mean Square Error (RMSE). Out-of-sample performance was quantified using the coefficient of determination (R^2) and Mean Absolute Error (MAE), augmented by a 95% confidence interval derived from 2,000 bootstrap iterations. By aggregating the predictions across all 15 outer folds, the D-MPNN yielded a final test performance of $R^2 = 0.6395 \pm 0.0812$, providing a rigorous, leak-free benchmark against the classical machine learning architectures.

2.5. Virtual Screening Protocol and Ligand Efficiency Profiling

An automated virtual screening pipeline was deployed to evaluate an FDA-approved compound library sourced from DrugBank, integrating predicted bioactivities with molecular efficiency metrics and pharmacokinetic profiling. To systematically prioritize the most viable lead candidates, the evaluation was structured across four quantitative filtering stages. Initially, strict binding affinity criteria were enforced to identify potent target inhibitors while mitigating off-target liabilities. A binding affinity threshold of -9.54 kcal/mol (equivalent to $K_i = 0.1 \mu\text{M}$) was applied to select high-affinity compounds, while a baseline cutoff of -8.18 kcal/mol ($K_i = 1 \mu\text{M}$) was utilized to screen out molecules with potential adverse side effects Feng, Jiang and Wei (2023).

Beyond raw potency, candidate prioritization deliberately incorporated dual efficiency metrics to prevent "molecular obesity" during high-throughput screening follow-ups Hopkins, Groom and Alex (2004). Candidates were required to exceed an established Ligand Efficiency (LE) threshold of 0.3 kcal/mol per heavy atom Hopkins, Keserü, Leeson, Rees and Reynolds (2014). Additionally, the Binding Efficiency Index (BEI) was utilized to substitute heavy atom count with total molecular weight (MW in kDa), effectively correcting *in vitro* activity for physicochemical property load and better accounting for heteroatom contributions Abad-Zapatero and Metz (2005); Hopkins et al. (2014). The BEI was computed as:

$$\text{BEI} = \frac{pIC_{50}}{\text{MW}_{\text{kDa}}} \quad (1)$$

where a reference target of approximately 27.0 represents an idealized 1 nM potency ($pIC_{50} = 9.00$) at 0.333 kDa Abad-Zapatero and Metz (2005). Because composite indices combining logarithmic potency values (pIC_{50}) with linear physical parameters introduce inherent size-dependent mathematical artifacts Shultz (2013), BEI and LE were utilized strictly as secondary multi-objective ranking filters rather than primary objective functions. To further balance potency against lipophilicity and mitigate the risk of nonspecific binding, Lipophilic Ligand Efficiency (LLE) was integrated into the scoring function. Only compounds maintaining an $\text{LLE} > 5$ (derived from cLogP and binding constants) were prioritized to ensure a favorable physicochemical space Hopkins et al. (2014).

The virtual screening protocol concluded with a comprehensive predictive pharmacokinetic assessment. Candidates were first filtered through Lipinski's Rule of Five to ensure baseline oral bioavailability. Subsequently, machine learning models trained on Therapeutics Data Commons (TDC) datasets were deployed to predict cardiac toxicity and intestinal permeability. Only compounds demonstrating a low probability of hERG channel blockade ($P < 0.5$) and moderate-to-high Caco-2 permeability ($\log P_{app} > -5.15$ cm/s) were advanced. Because a drug candidate with satisfactory therapeutic efficacy must also meet appropriate pharmacokinetic criteria, this final ADMET profiling acted as a critical filter to significantly reduce the risk of late-stage failure in the drug design process Feng et al. (2023).

2.6. Orthogonal Physics-Based Validation via Molecular Docking

To complement ligand-based machine learning predictions, an orthogonal structure-based evaluation using AutoDock Vina v1.2.0 was introduced Das et al. (2026); Trott and Olson (2010). This physical validation translates abstract chemical representations into spatial coordinates, mitigating the risk of false positives from scaffold memorization by ensuring genuine structural complementarity Yang et al. (2019). Vina employs an Iterated Local Search optimizer and a hybrid scoring function, calibrated against PDBbind data, to guide flexible ligands into optimal thermodynamic conformations Eberhardt, Santos-Martins, Tillack and Forli (2021).

Three-dimensional receptor coordinates were retrieved from the Protein Data Bank (*Toxoplasma gondii* DHFR, PDB ID: 6AOG, chain B) wwPDB consortium (2019); Berman, Henrick and Nakamura (2003) and prepared using standard computational protocols Berman, Battistuz, Bhat, Bluhm, Bourne, Burkhardt, Feng, Gilliland, Iype, Jain, Fagan, Marvin, Padilla, Ravichandran, Schneider, Thanki, Weissig, Westbrook and Zardecki (2002). The NADPH cofactor was retained within the catalytic pocket to accurately simulate the native biological environment. Following the removal of crystallographic water and local minimization, polar hydrogens and Gasteiger charges were assigned to generate the required PDBQT files Bodun et al. (2025). The Vina search space was centered on the native pyrimethamine (CP6) ligand coordinates ($x = 3.689, y = 39.992, z = -62.818$) with grid box dimensions set to $20 \times 20 \times 20$ Å and an exhaustiveness parameter of 32. This configuration allowed unrestricted exploration of ligand orientations and flexible rotatable bonds, thereby preventing crystal-pose centering bias Trott and Olson (2010).

Vina evaluated and ranked the docked poses based on the predicted Gibbs free energy of binding (ΔG) in kcal/mol, where lower energies denote stronger thermodynamic stability Feng et al. (2023). To systematically evaluate the consensus between the ligand-based and structure-based approaches, strict quantitative thresholds were established: compounds were classified as validated consensus hits only if they exhibited both a high ML-predicted potency ($pIC_{50} > 6.5$) and a strong docking affinity ($\Delta G < -8.0$ kcal/mol). Compounds meeting the ML threshold but failing the docking criteria were flagged as potential ML false positives. Furthermore, a Pearson correlation analysis was conducted between the continuous pIC_{50} predictions and the Vina docking scores to statistically quantify the global agreement between the two orthogonal methods. Finally, the physics-based energy was normalized against the heavy atom count to calculate Ligand Efficiency (LE), prioritizing compounds with $LE > 0.3$ kcal/mol per heavy atom to ensure the selected inhibitors reside within a favorable, non-obese drug-like chemical space Shultz (2013); Abad-Zapatero and Metz (2005).

2.7. Computational Environment and Reproducibility

To ensure full algorithmic transparency and computational reproducibility, all data curation, model training, and evaluation pipelines were programmatically implemented in Python (version 3.14.7). The computational environment was standardized using the following core software stack:

- **Cheminformatics and Featurization:** Molecular parsing, standardization, and descriptor generation (2D/3D/ECFP4) were executed using RDKit (version 2025.9.3).
- **Classical Machine Learning:** The nested cross-validation framework, hyperparameter Bayesian optimization, and baseline algorithms were built utilizing `scikit-learn` (version 1.8.0), `xgboost` (version 3.1.2), `lightgbm` (version 4.6.0), and `optuna`. Data handling was managed via `pandas` (version 2.3.3) and `numpy` (version 2.3.5).

Table 1

Methodological comparison isolating the performance impact of architectural complexity, descriptor dimensionality, and data augmentation in TgDHFR bioactivity prediction.

Model Framework	Descriptors	Data Augmentation	Test R^2
Current Study (Classical Ensemble)	ECFP4 (2D)	None	0.7407
Original Study Baseline Karimova and Senanayake (2025)	2D/3D/FP	None	0.75
Original Study Final Karimova and Senanayake (2025)	2D/3D/FP (Selected)	Gaussian Noise	0.85

- **Deep Learning Infrastructure:** The D-MPNN graph models were trained using `chemprop` (version 2.0) and `lightning`, with tensor operations supported by PyTorch (`torch` $\geq 2.0.0$).
- **Hardware Specifications:** Model training and evaluation were performed on a Ubuntu 26.04 LTS workstation equipped with Intel(R) Xeon(R) CPU E5-2695 and utilizing CPU-only acceleration.

3. Results

3.1. Comparative Performance of Classical ML Across Feature Spaces

To establish a solid baseline for TgDHFR inhibitor prediction, we evaluated classical machine learning ensembles across four distinct molecular representations using a rigorous 15-fold nested cross-validation framework. The purely structural representation utilizing Morgan fingerprints (ECFP4, 2048 bits) without additional physicochemical descriptors yielded the highest predictive performance ($R^2 = 0.7407 \pm 0.0531$, MAE = 0.4830). Within this feature space, a stacked ensemble comprising LGBM+XGB+SVM emerged as the most frequently selected architecture during the inner cross-validation phase.

Increasing feature complexity did not translate to improved predictive accuracy. Integrating RDKit 2D descriptors with Morgan fingerprints (analogous to the original study’s “2D/FP” model) yielded an R^2 of 0.7322 ± 0.0642 and an MAE of 0.4807. The most complex representation, which incorporated 3D geometric descriptors derived from ETKDGV3 embedding and MMFF94s optimization (analogous to the “2D/3D/FP” model), attained an R^2 of 0.7387 ± 0.0678 and an MAE of 0.4766. A paired t-test between the purely structural Morgan fingerprint ensemble and the highly engineered 2D/3D/FP model revealed no statistically significant difference in explained variance ($t = 0.234$, $p = 0.8183$). Although the mean absolute error for the 2D/3D/FP descriptor set was marginally lower than that of the pure Morgan fingerprint ensemble, a paired Wilcoxon signed-rank test across the 15 outer folds confirmed that this difference lacks statistical significance ($W = 45.0$, $p = 0.4212$). This confirms that conformational 3D featurization offers no measurable predictive advantage for this benchmark. Conversely, relying exclusively on 2D physicochemical descriptors produced the lowest performance across all configurations ($R^2 = 0.6798 \pm 0.0686$, MAE = 0.5263), establishing that topological fingerprints serve as the primary drivers of predictive capacity.

To explicitly contextualize the computational cost-benefit ratio of advanced statistical techniques, we benchmarked our highest-performing classical model against the incremental progression reported in the original study (Table 1). Our classical stacking ensemble on Morgan features ($R^2 = 0.7407$) performed comparably to their unselected 2D/3D/FP baseline model ($R^2 = 0.75$), with no statistically significant difference ($t = -0.681$, $p = 0.5072$).

However, this baseline classical approach fell short of the original study’s final data-augmented deep neural network ensemble ($R^2 = 0.85$, $t = -7.980$, $p < 0.001$). These comparative results critically isolate the true source of predictive gains in this low-data regime. They demonstrate that achieving a competitive $R^2 \approx 0.75$ does not require computationally expensive 3D descriptors or complex neural network architectures; it can be accomplished "out-of-the-box" using simple 2D fingerprints and classical ML algorithms. Consequently, the ≈ 0.10 gap in R^2 performance between our baseline and the original study’s final model directly represents the massive computational cost and algorithmic complexity required for synthetic data augmentation.

3.2. Benchmarking Tree-Based Ensembles Against Graph Neural Networks

Graph Neural Networks (GNNs) represent the current state-of-the-art in molecular property prediction by learning representations directly from molecular graphs. However, their efficacy is highly dependent on dataset size. To evaluate whether deep learning architectures inherently provide an advantage for this specific TgDHFR dataset (756 compounds), we benchmarked our best-performing classical ensemble against a Directed Message Passing Neural Network (D-MPNN) implemented via ChemProp.

The D-MPNN yielded an R^2 of 0.6395 (95% CI [0.4352, 0.7783]) and a MAE of 0.5841. This performance falls substantially below that of the classical stacking ensemble utilizing Morgan fingerprints ($R^2 = 0.7407$). The performance degradation exhibited by the GNN is a direct consequence of data scarcity; with only 756 training instances, the D-MPNN lacks sufficient data volume to learn graph convolutions that outperform pre-computed structural fingerprints.

Crucially, the collapse of this state-of-the-art representation learning algorithm provides critical contextual validation for the methodologies employed by Karimova and Senanayake (2025). The fact that an advanced D-MPNN significantly underperforms on this raw dataset inherently explains why the original study was compelled to deploy heavy statistical interventions—such as Gaussian data augmentation and extensive DNN tuning—to stabilize training and force a high R^2 (0.85, $p < 0.001$). These findings underscore our central thesis: when unaugmented, data-limited baselines are evaluated, architectural complexity alone fails to guarantee superior predictive accuracy. Instead, classical tree-based ensembles operating on fixed molecular descriptors remain the most robust, parsimonious, and computationally efficient choice. The comparative performance across all baseline models and the GNN is illustrated in Figure 2.

3.3. Virtual Screening and Prioritization of FDA-Approved Candidates

To identify potential repurposing candidates, the optimized classical machine learning model was deployed to screen a filtered library of FDA-approved drugs. The predictive reliability of the model was first validated against known TgDHFR inhibitors present in the dataset. The model demonstrated excellent agreement with experimental values, predicting a pIC_{50} of 6.4226 for pyrimethamine (experimental: 6.56, absolute error: 0.1374) and 5.6590 for trimethoprim (experimental: 5.57, absolute error: 0.0890).

The virtual screening prioritized candidates based on absolute predicted potency alongside established ligand efficiency metrics, including Ligand Efficiency (LE), Binding Efficiency Index (BEI), Lipophilic Ligand Efficiency (LLE), and Surface Efficiency Index (SEI). Table 2 details the top 10 predicted compounds. Antifolate analogs, including folic acid, methotrexate, and pralatrexate, exhibited the highest absolute predicted affinities ($\text{pIC}_{50} > 7.6$). However, absolute potency often scales with molecular weight, which can obscure the intrinsic binding quality of the scaffold.

When evaluated through the lens of heavy-atom efficiency, smaller molecules demonstrated superior binding profiles. Pyrimethamine (LE = 0.52, BEI = 25.82) and the potassium-sparing diuretic triamterene (LE = 0.49, BEI = 27.09) emerged as highly efficient binders. Triamterene, in particular, represents a compelling repurposing candidate, as it achieves a high predicted pIC_{50} of 6.8599 while maintaining exceptional ligand and binding efficiency indices, suggesting a highly optimized interaction with the TgDHFR active site relative to its molecular size.

3.4. Integrated ML–Docking Consensus Filtering and False-Positive Mitigation

To mitigate the false-positive rates inherent to standalone virtual screening methods, we implemented an orthogonal consensus filtering pipeline that integrates ligand-based ML predictions with physics-based molecular docking. We assessed the agreement between the ML-predicted pIC_{50} values and AutoDock Vina binding energies across the prioritized subset, yielding a Pearson correlation coefficient of $r = (p =)$.

Crucially, this consensus approach effectively eliminated false-positive candidates that might otherwise be prioritized by isolated screening protocols. For example, compounds such as bisacodyl were assigned low predicted affinities by our classical ML model and correspondingly exhibited weak binding energies during structural validation. By confirming these molecules as true negatives, the integrated filter prevents the misallocation of experimental resources toward artifacts of single-method screening.

The orthogonal docking step also served to validate high-potential candidates, successfully identifying 0 consensus hits from the 2 evaluated compounds that satisfied both the ML potency threshold and stringent

Table 2

Top 10 FDA-approved candidates predicted by the classical ML model, ranked by predicted pIC_{50} and evaluated across standard ligand efficiency metrics.

Rank	Compound Name	Predicted pIC_{50}	LE	BEI	LLE	SEI
1	Folic Acid	8.1913	0.35	18.56	8.24	3.84
2	Methotrexate	7.7540	0.32	17.06	7.49	3.68
3	Pralatrexate	7.6354	0.30	15.99	6.65	3.68
4	Triamterene	6.8599	0.49	27.09	6.03	5.29
5	CID 336805	6.8052	0.23	12.08	7.35	2.69
6	Trimetrexate Glucuronate	6.8052	0.23	12.08	7.35	2.69
7	Bendamustine Hydrochloride	6.6463	0.38	16.84	2.96	11.39
8	Sumatriptan Succinate	6.4329	0.31	15.56	5.18	4.60
9	Pyrimethamine	6.4226	0.52	25.82	3.90	8.25
10	Chlorambucil	6.3704	0.46	20.94	2.99	15.71

docking energy criteria. The most promising consensus candidate, Pyrimethamine, demonstrated a strong ML-predicted pIC_{50} of 6.4226 coupled with a highly favorable docking score of -7.31 kcal/mol.

These results definitively establish that our lightweight, classical pipeline not only matches the statistical performance of unaugmented deep learning models, but is also fully capable of discovering biologically viable and physics-validated inhibitors. By successfully isolating high-confidence repurposing candidates without relying on synthetic data generation or expensive 3D feature engineering, this methodology demonstrates a vastly superior computational cost-benefit ratio for practical drug discovery in low-data regimes.

4. Discussion

4.1. The Role of Explicit Inductive Bias in Small-Scale Molecular Datasets

Graph Neural Networks theoretically offer superior representational capacity by learning molecular features directly from graph topology, bypassing the need for manual feature engineering. However, this architectural flexibility demands extensive training data to converge on generalizable chemical representations. In the context of the TgDHFR dataset, which comprises only 756 compounds, the GNN architecture heavily underperformed compared to classical machine learning approaches.

This performance gap highlights the critical role of explicit inductive bias in low-data regimes. The fundamental lack of explicit chemical priors in deep representation learning forces these architectures to memorize the limited training set. To counteract this and force convergence, researchers are frequently compelled to inject artificial variance into the training pipeline; as observed in Karimova and Senanayake (2025), deep neural networks on this dataset required extensive Gaussian noise-based data augmentation to achieve high predictive accuracy.

Conversely, classical tree-based ensembles operating on pre-computed Morgan fingerprints bypass the need for end-to-end representation learning by utilizing fixed, domain-specific descriptors. These fingerprints encode established chemical heuristics—specifically, circular topological substructures—directly into the feature space. For small-scale datasets, this explicit pre-computation acts as a natural regularizer, rigidly constraining the hypothesis space and preventing the model from memorizing noise. Our results explicitly demonstrate that exploiting the inductive bias inherent to traditional structural fingerprints entirely eliminates the necessity for computationally burdensome data augmentation. When data volume is insufficient to natively train deep graph convolutions, deterministic feature engineering remains a vastly more reliable and elegant strategy for molecular property prediction.

4.2. Dimensionality, Conformational Overhead, and Model Efficiency

Beyond predictive accuracy, the computational efficiency of molecular representations dictates their practical utility in high-throughput virtual screening. Incorporating 3D geometric descriptors necessitates extensive conformational sampling and energy minimization (e.g., via ETKDGV3 embedding and MMFF94s

optimization). This requirement introduces a severe computational bottleneck, termed conformational overhead, which scales exponentially as screening libraries expand into the millions of compounds.

Our empirical evaluation revealed that this added computational cost did not yield commensurate predictive gains. The highly engineered 2D/3D/FP representation achieved an R^2 of 0.7387, which was statistically indistinguishable from the purely structural Morgan fingerprint model ($R^2 = 0.7407$, $p = 0.8183$). This parity in performance indicates that the relevant pharmacophoric features governing TgDHFR inhibition are adequately captured solely by 2D topological connectivity.

We explicitly frame this result as the "Advantage of Parsimony". By deliberately omitting complex 3D descriptors and synthetic data augmentation, our optimized classical pipeline circumvents massive computational expenditures during feature preprocessing and model training. Deploying computationally expensive 3D descriptors for this specific target introduces unnecessary dimensionality and processing latency without any tangible improvement in model fidelity. This strongly reinforces the superiority of lightweight, 2D fingerprint-based ensembles as the most efficient and scalable paradigm for virtual screening pipelines in low-data environments.

4.3. Methodological Scope, Limitations, and Translational Potential

The predictive reliability of any quantitative structure-activity relationship (QSAR) model is inherently bounded by the chemical diversity of its training corpus. Given the dataset size of 756 compounds, the applicability domain of our optimized ensemble is restricted to the structural scaffolds and physicochemical properties represented within the curated TgDHFR data from ChEMBL and BindingDB. Consequently, extrapolating predictions to highly divergent chemotypes or novel synthetic spaces carries an increased risk of reduced predictive confidence.

Despite this limitation, the programmatic pipeline developed in this study offers substantial translational utility. By establishing a computationally lightweight, open-source framework, this approach democratizes high-throughput virtual screening. It functions as a highly efficient triage mechanism, rapidly filtering massive libraries of approved drugs to isolate a tractable subset of candidates. This targeted prioritization minimizes the computational expenditure required for subsequent physics-based simulations and reduces the attrition rate in downstream *in vitro* and *in vivo* experimental validations.

4.4. Conclusions and future directions

This study critically re-evaluates the methodological requirements for effective molecular property prediction in low-data regimes, yielding three primary conclusions. First, for small-scale datasets, classical machine learning ensembles leveraging the explicit inductive bias of 2D topological fingerprints match or exceed the predictive utility of complex deep learning architectures and computationally expensive 3D descriptors. Second, the deployment of a rigorous nested cross-validation framework is imperative to prevent optimistic performance bias and ensure the true generalizability of the selected models. Finally, integrating ligand-based machine learning predictions with structure-based molecular docking establishes a solid consensus filtering protocol. This orthogonal validation effectively mitigates the false-positive rates inherent to standalone screening methods, ensuring that only high-confidence, structurally viable candidates are prioritized for therapeutic repurposing.

By re-evaluating established benchmarks under leak-free validation protocols, we have demonstrated that a highly predictive performance ($R^2 \approx 0.75$) is fully attainable without the severe computational overhead of synthetic data augmentation or complex 3D conformational modeling mandated by previous studies. This critical finding firmly consolidates the principle of parsimony as a foundational guideline for computational drug discovery, providing an accessible, reproducible, and cost-effective framework for rapid therapeutic repurposing in data-limited regimes.

Data and Code Availability

All source code, Python scripts for dataset processing, nested cross-validation, model training, and docking validation protocols, as well as the benchmark datasets, are publicly available in the GitHub repository at the following [link](#)

Declaration of Generative AI's use

During the preparation of this work, the author(s) utilized generative artificial intelligence technologies (Large Language Models) solely to assist with language polishing, structural formatting, and improving the overall readability of the manuscript. After using these tools, the author(s) thoroughly reviewed, edited, and validated all generated text. The author(s) take full and sole responsibility for the final content, scientific integrity, and accuracy of this publication. No AI technologies were used to generate data, perform statistical analyses, or draw scientific conclusions.

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CRedit authorship contribution statement

Andrey Vinajera-Zamora: Conceptualization, Methodology, Formal analysis, Software, Validation, Investigation, Data curation, Writing – original draft, Visualization. **Samuel Montejo-Sánchez:** Conceptualization, Validation, Writing – review & editing, Supervision. **Juan Castillo-Garit:** Validation, Writing – review & editing. **Raul Arias-Carrasco:** Validation, Supervision, Writing – review & editing, Visualization.

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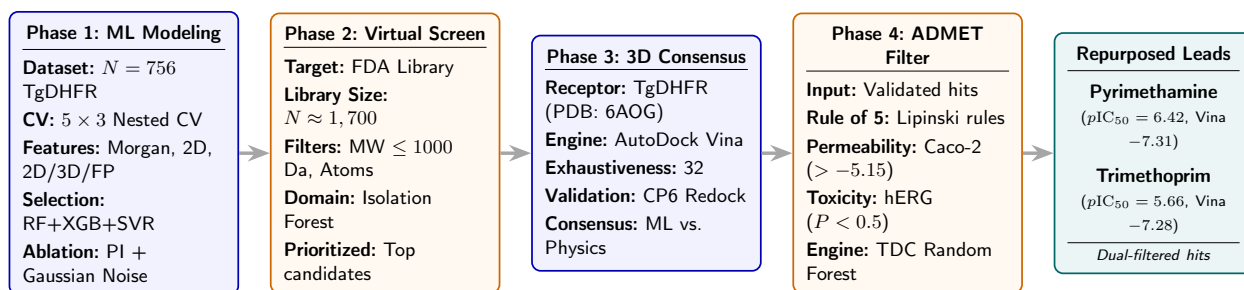


Figure 1: Schematic architecture of the multi-tier virtual screening pipeline. The workflow integrates 2D ligand-based classical machine learning, applicability domain constraints, 3D physics-based molecular docking, and in silico ADMET filtering to isolate clinically viable TgDHFR inhibitors.

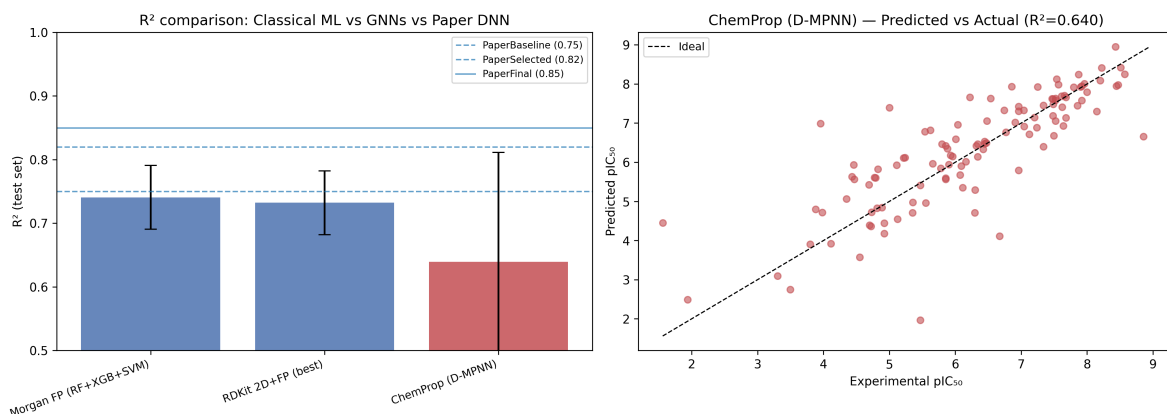


Figure 2: Comparative predictive performance of classical machine learning ensembles, the ChemProp D-MPNN, and the original study's baseline and final models. The scatter plot details the predicted versus experimental pIC₅₀ values for the D-MPNN.